

II

A/

Security Attack $\rightarrow$ It is a planned action to break the security and gain unauthorized access for data which is not for the public.

Security Vulnerability $\rightarrow$ It is the incompleteness of security measures and there is an unfair advantage within the security measures for the attacks to gain unauthorized access.

B/

Packet Cellification in queuing and scheduling.

The packets need to be in a proper format for queuing and scheduling. The process is called cellification.

This cellification is perfect because it orders and coordinates with every thing to make a proper cell. After cellification, all the cells are sent to queuing and scheduling.

There are many methods like $\rightarrow$

- First come First serve
- Weighted Round Robin

C] Trojan horse

A Trojan horse is a virus hidden under a non suspicious file. The file may seem normal and also function as normal. But within the system its virus will spread and infect the machine. This name is originated from the war era as they used to send horse sculptures as a gift but it would contain army men inside it. So after accepting it the men would attack and destroy it from within.

D] Symmetric

- Symmetric encryption
- Same key for encryption and decryption
- Need initial trust to give public key
- Not that safe

asymmetric.

- Asymmetric encryption
- Diff key for encryption and decryption
- Public key can be given without great fear.
- Comparably safer

E]

Digital Signature

A Digital Signature is a piece of software which has a generated signature which can't be duplicated. It can't be duplicated so it is generally used as a form of authentication/verification that it has been done by a certain specific user, used for verification.

F] Realm in Kerberos.

In Kerberos there are 2 parts

The user.

- The Key Distribution Centre.

Realms in Kerberos is so that each Kerberos are in blocks of its and cross platform communication can occur. The user pings the KDC and it acknowledges with id a key to authenticate the user. Then the user can ping another Kerberos and it will cross verify it it has a key with the other Kerberos.

2

A Pro-active password checking

It is suggested to not reuse a password for anything. Once this was better to not use that same password again. There can be variation but not the exact same thing. If any hacker hacks and takes your password, he will have access to all the accounts with the same password.

Each password must contain

- Capital
- Small
- Number
- Symbol

Password Manager is essential for every one. Your passwords must be kept safe and not to be known by others. You can use Google password manager if you have lost control of your passwords.

Stay safe,

B

SNMP

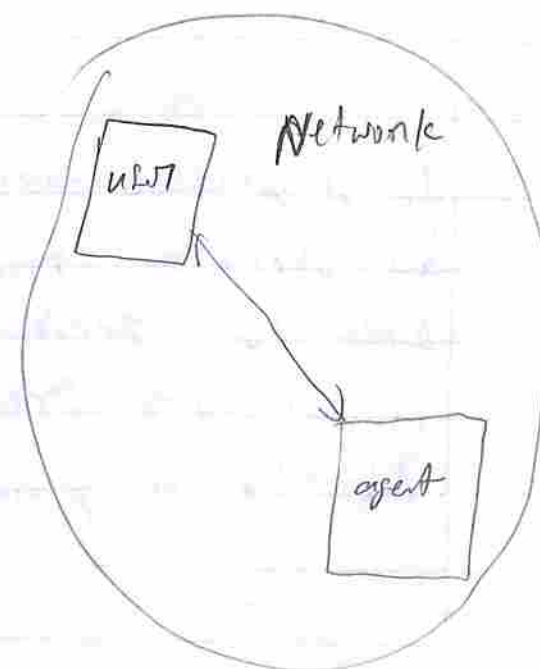
Simple network management protocol.

It contains: ~~ATB~~

- MIB → management interface Database
- SMi → Smart Management interface.
- Snmp agent → agent.

SNMP is a simple network management protocol, it is used to manage a simple network. It has agents which do its work for you.

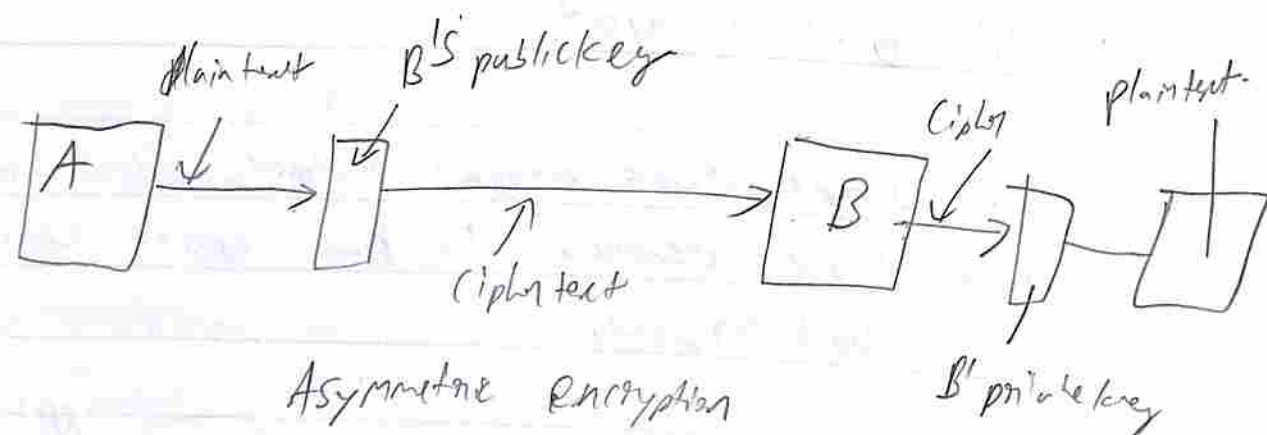
The user is connected to a network and tries to ping the agent. The agent then sends a response at the ping and notifies the user.



C

Mutual authentication,

In mutual authentication, both the users authenticate each other to verify who they are communicating with.

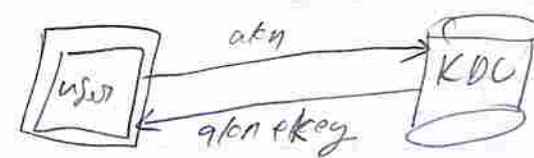


Asymmetric Encryption

In Asymmetric Encryption there are 2 keys used for encryption and decryption respectively. The sender encrypts the data with the receiver's public key. The now ciphertext is sent to the receiver, and the receiver decrypts the ciphertext to plaintext with help of his private key.

D KDC

Key Distribution Centre is a all public directory of all the keys. You can add your new keys to it and it will be known to all. It gets renewed every 2 years.



It can be duplicated because everything is public and known, even if it is tried, we can cross verify and find out the original.

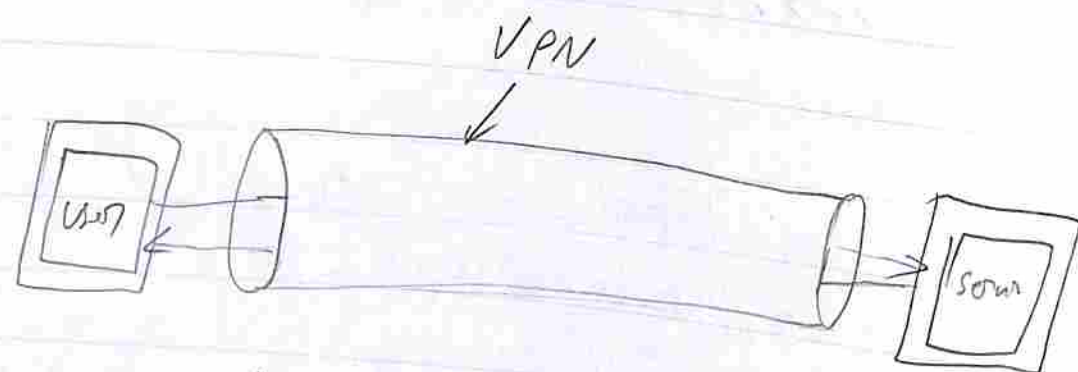
A great example of this type public library is DNS.

3

A VPN

Virtual private network...

To keep your network more secure we can use a VPN. So that your public IP will be hidden and changed also.



VPN acts like a tunnel to our public IP address. It hides and covers our information so we can be safe.

There are many VPNs like

• NordVPN

• Not shield VPN

B) Possible attacks on routers.

Routers are a network device and are a common target to attack. They try to ~~break~~ and gain unauthorized access to the data. A few common examples are

• Router Flooding (DoS)

They flood the router with an overload of requests so that the router has burnt itself out unable to function.

A common countermeasure for this is rate limiting, so that the requests are limited.

• Router table poisoning.

They fill the router table with false information.

A common counter measure is to have access control

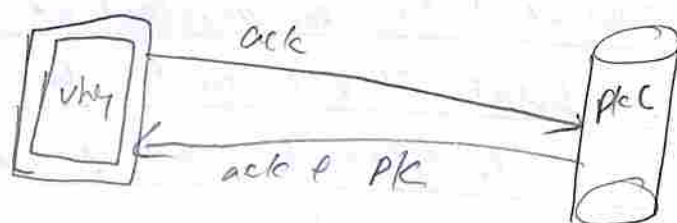
51

A1

Public Key Certificate

A Public Key Certificate is automatically generated and can't be manually generated.

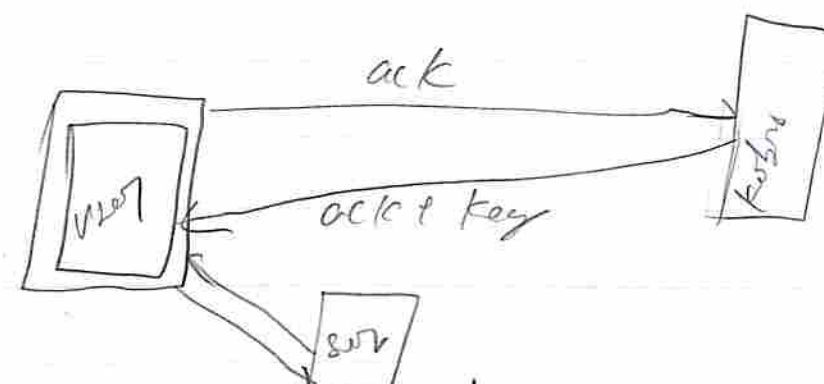
It is used to authenticate and verify the public key.



These Public Keys can also be in a database like Key Distribution Centre (KDC)

Public Key Certificate is very important and can't be duplicated

B1 Kerberos



The Kerberos authentication scheme is that there are 3 components

- User
- Serv
- Authentication Serv

The user picks the authentication server and is authenticated by that server. Then he can access the other server. The other server can cross verify with the authentication server.

This is how Kerberos works!

